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Student Drop-out from German Higher Education Institutions

Ulrich Heublein

Introduction

Student drop-out is currently an important issue in higher education institutions in Germany. The virulence with which politics pose questions about drop-out from higher education is due to current labour market trends. Despite a rise in student numbers, there is a severe lack of academically-qualified persons in a whole range of industries. This applies more particularly to graduates in natural and engineering sciences. Reducing the drop-out rate is an effective way of increasing the respective labour supply.[1] These trends are associated with the recent performance-oriented, indicator-based and formula-linked methods of funding both to higher education institutions and within higher education institutions themselves. As a complementary value, the success rate is a main performance indicator in many models of fund distribution. Importance is also given to success or drop-out rates in the quality assurance of teaching, the evaluation of teaching practices, or the accreditation of degree programmes. This is relevant, notwithstanding the fact that central aspects of the issue of drop-out from higher education require a more in-depth explanation. The question about whether high drop-out rates are due to specific conditions in higher education institutions or to factors that are not influenced by the particular higher education institution must remain partially open. Too often, the political debate is limited to the questionable determination of individual academic success through local studying conditions.

As a reaction, the empirical research on students and higher education in Germany poses three main questions in the study of drop-out from higher education institutions: first, there is the question concerning the statistical data on the extent of student drop-out in the different types of higher education institutions. This is linked to problems concerning the definition and measurement of drop-out from higher education. Secondly, the research aims at clarifying the individual, institutional and social causes. Closely linked to this is the third question that focuses on above-average drop-out dangers within specific at-risk groups.

Determination of Drop-out Rates

Vague definitions of student drop-out and the selection of different reference points proved to be great obstacles for the discourse on student drop-out for years. Even today, especially in the debate about higher education policy, there is a statistical confusion due to a lack of differentiation between the types of fluctuation in the higher education system. Without a clear conceptual distinction between student drop-out and forms of student mobility such as interruption in studies, change in degree programme or subject, change of institution, change of type of higher education, or going abroad, it is not possible to precisely capture this premature study termination. The research on higher education offers the terms shrinkage and shrink balance for the mobility processes in the higher education system. As a measure of shrinkage, the shrink rate can be defined at the level of the

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degree programmes, the subjects, the institutions and the overall system. At the level of the degree programme, it includes all first-year students of any one year who left the programme without obtaining an academic degree, regardless of whether this is due to a change in degree programme within the higher education institution, a change to a different institution, or drop-out. However, a degree programme does not only lose students within the course of a year, but frequently gains others through successful inward migration. In most cases, the higher education institutions are unable to distinguish between drop-out and other shifts, since they do not have any information on the academic career of their students after their deregistration.

The empirical research on higher education defines student drop-out as a special type of shrinkage that only includes students who leave the higher education system without obtaining a (first) degree and do not complete their studies at a later stage.[2] This narrow understanding of the term has gained ground in research (Schröder-Gronostay & Daniel, 1999; Ziegele, 1997; Heublein, Schmelzer & Sommer, 2005). Hence, the change in degree programme or subject, interruption in studies and the change of institutions are different types of shrinkage. There are several ways of determining the frequency of student drop-out. The best is a statistic on the course progression with which the study status of each student is recorded every semester. Germany currently lacks the legal preconditions for this.[3] While there are data on deregistered students collected by higher education institutions that determine the number of drop-outs, the statistics do not offer the possibility to do so, either at the institutional level or at the level of the higher education system. Since students are not obliged to state the reason for their withdrawal, the reasons for deregistrations remain unknown to the higher education institutions.

Representative panel studies that trace groups of first-year students over the entire course progression is another way of determining drop-out rates. In Germany, the National Educational Panel Study (NEPS) and the Studienberechtigtenpanel of the German Centre for Research on Higher Education and Science Studies (DZHW) are the most important. However, neither these nor other panel projects have succeeded in reaching sufficient numbers of random samples of drop-outs so far.[4] The difficult problem arises of how to address the participants, especially those who abandoned their studies during the first inquiry, so as to motivate them to collaborate. Since neither official statistics nor panel studies are able to measure the extent of drop-out in Germany, statistical estimations that lead to an approximation of the real value must be used. There are many approaches to these estimations in the empirical research on higher education. In Germany, two approaches that offer valid drop-out rates were established recently.

A differentiated approach to the calculation of academic success rates as a counterpart to the share of drop-outs is adopted by the Federal Statistical Office (Hetmeier *et al.*, 2008; Scharfe, 2010). In the analysis, a group of first-year students is considered. Students' status at a specific moment, i.e. if they completed their studies, if they still attend a higher education institution, or if they deregistered without an academic degree, is determined on the basis of the data of the examination and student statistics. The timing for this classification is chosen in such a way that the share of students that is still attending a higher education institution is not greater than 20%. The likelihood of drop-out is estimated among these students. The allocation of first-year students to the graduates is possible via

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a statistical link that is composed of invariant attributes of the student statistics. About four-fifths of all students can be recorded in this way. From the relation of the first-year students of any one year and their assigned graduates, including the estimated number of graduates as a proportion of students who still attend a higher education institution, the academic success rate is then estimated for the examined group of any one year. With this approach, the Federal Statistical Office obtains an estimated academic success rate of about 76% for the groups of first-year students from 1997 to 2002, which equates to a drop-out rate of 24%.

The second method was developed by the DZHW (formerly Hochschul-Informations-System (HIS)) (Heublein *et al.*, 2012). It is a retrospective comparison between graduates and first year-students of any one year. However, the relation is not established through the average length of studies, but with the help of a newly-formed group in which all groups of first-year students are included with a weight that is equal to their respective share in the examined group of graduates. Several adjustments must be made to construct this 'virtual' group of first-year students. On the basis of the periodic survey of first-year students by the DZHW, the numbers of first-year students are reduced through double registrations and registrations in a second degree course which equates to a range between 2% and 6%. The differentiated student drop-out rates can only be determined if the student behaviour in the change of subjects is considered and the graduates are mathematically attributed to the group of first-year students to which they belonged at the time of their undergraduate registration. This calculation is carried out with the help of a change of matrix that depends on the findings of the representative survey of graduates conducted by the DZHW. While the Federal Statistical Office proceeds from a single group of first-year students of any one year and the rates solely refer to this specific group, the DZHW proceeds from a group of graduates and establishes the drop-out extent for two to three consecutive groups of first-year students with the help of a newly-formed corresponding group of first-year students. The Federal Statistical Office obtains drop-out rates of 24% from 2000 to 2004. In a differently defined method, the DZWH calculated a drop-out rate of 21% to 24% for the same groups of first-year students. Furthermore, hardly any disparities in trends can be seen in the two approaches in the different subject groups. For this reason and also because the Federal Statistical Office did not submit drop-out figures for the current groups of first-year students, the following illustrations of drop-out since the shift to successive study programmes are limited to the rates determined by the DZHW.

Drop-out Extent

The introduction of the bachelor programme led to a rise in student drop-out in German higher education institutions. While the data of groups of first-year students in diploma study programmes between 1999 and 2008 fluctuated within a margin of 21% to 25%, the student drop-out rates rose to 30% among students in the first bachelor degree programmes in 2001–2002 (Heublein *et al.*, 2014). These relatively high rates were caused by the transition and conversion difficulties linked to the bachelor programme. However, there was no significant reduction in drop-out for the following groups of first-year students either. The latest data for the groups of first-year students are from 2008 and 2009. 28 out of 100 bachelor students still complete their studies without obtaining a first academic degree (Figure 1).

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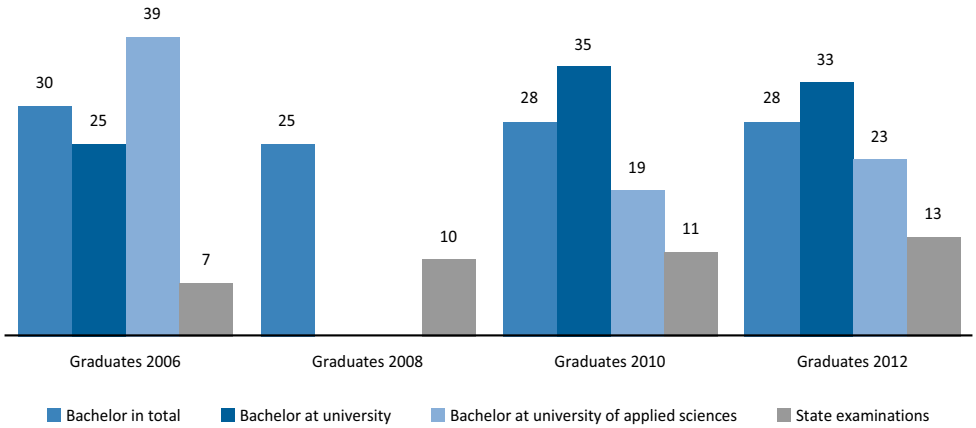

* The student drop-out rates in the different types of degree programmes relate to differing cohorts of first-year students

DZHW 2014

FIGURE 1. Development of the drop-out rate by type of degree*

Within this framework, there are clear differences between bachelor programmes at universities and those at universities of applied sciences. Although the universities were able to point to only a slight increase in drop-out (25%) at the outset of the conversion to bachelor programmes, this situation has changed. With drop-out rates of 33%, the rates of the bachelor programme at universities are clearly above the corresponding rates of universities of applied sciences. During the transition to successive study programmes, this development was caused by the introduction of the bachelor programme in those faculties whose teaching culture — e.g. in business administration — traditionally showed only moderate drop-out rates. When the bachelor programme was introduced in additional subjects, especially in natural sciences and engineering, the drop-out rate rose to an above-average level. There was a quicker transition to bachelor programmes at universities of applied sciences. From the start, the high drop-out rates that characterised natural science and engineering subjects were included. At universities and at universities of applied sciences it was very difficult for these subjects to develop the curricular prerequisites and appropriate study requirements for this new type of programme. Hence, with the introduction of the bachelor degree programme at universities of applied sciences, there was a strong rise in drop-out rates to 39%. Through counter measures, it stood at 23% for the groups of first-year students in 2008–2009.

In addition, specific developments of subject groups and fields of study exert an influence on drop-out rates. At universities, mathematics, natural sciences and engineering register a particularly high loss of students of 39% and 36% respectively in their bachelor degree programmes. However, varying trends can be found. Thus, high drop-out rates of roughly 40% are found in mathematics, computer science, physics and chemistry, as well as in mechanical, electrical and civil engineering. Biology, geography and architecture have below-average rates. Similarly, there are high drop-out rates in the humanities and linguistics and low drop-out rates in educational sciences, psychology, law economics and social sciences. The agricultural, forestry and nutritional sciences fall within the average rates (Figure 2).

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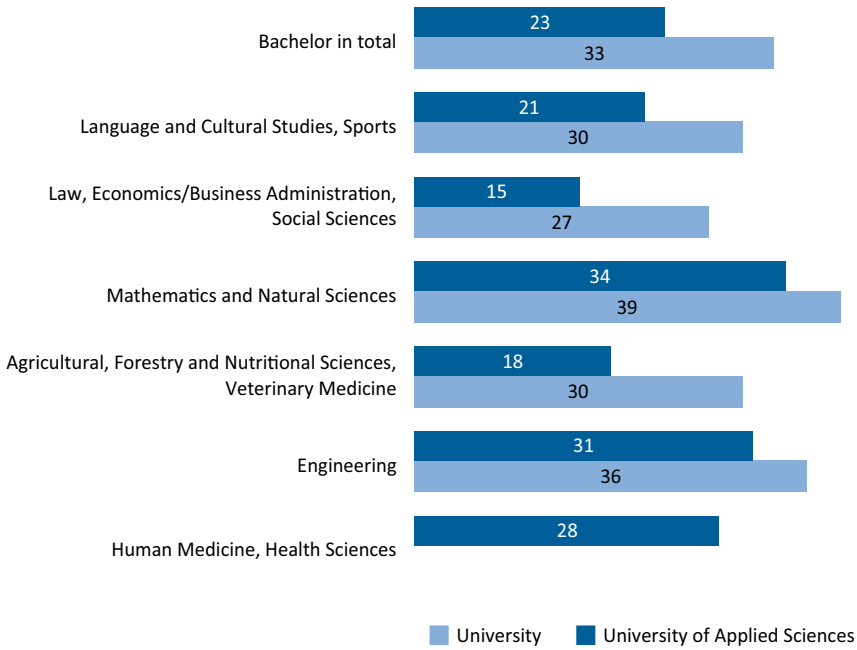

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FIGURE 2. Drop-out rates in bachelor degree programmes at universities and universities of applied sciences, by subject groups

At universities of applied sciences, the highest drop-out rates are found in mathematics, natural sciences and engineering where they were about 10 percentage points higher than the average. Architecture and certain engineering disciplines other than mechanical, electrical and civil engineering are exceptions. Whereas in the last few years newly-introduced jurisprudential programmes have shown comparatively high drop-out rates of 27%, the share for business administration and especially the social system and welfare studies is very low (Figure 3).

Drop-out is much lower in the master degree programme. Only 11%[5] of students who chose this programme at university in 2010 did not obtain a degree. At universities of applied sciences, this proportion is only 7%. Despite a lack in empirical evidence concerning academic success, the strong selectivity during the bachelor programme and a performance-oriented admission to a master programme may account for the lower number of drop-outs, although there are considerable differences between faculties. Particularly noticeable is the low number of drop-outs in mathematics and natural sciences. This is certainly an outcome of the performance-related selection in the previous study phase, but also of a lack of vocational attractiveness of a bachelor's degree in these fields. In contrast, drop-out in the business administration master programme has far fewer vocational consequences.[6] Hence, the assumption about commencing studies in order to initiate a smooth transition to a professional occupation seems to be legitimate for some students in a master programme.

At German universities, drop-out remains low in subjects which lead to a state examination such as medical and teacher training programmes, whose drop-out rates based on the groups of first-year students from the years 2006 to 2008 are

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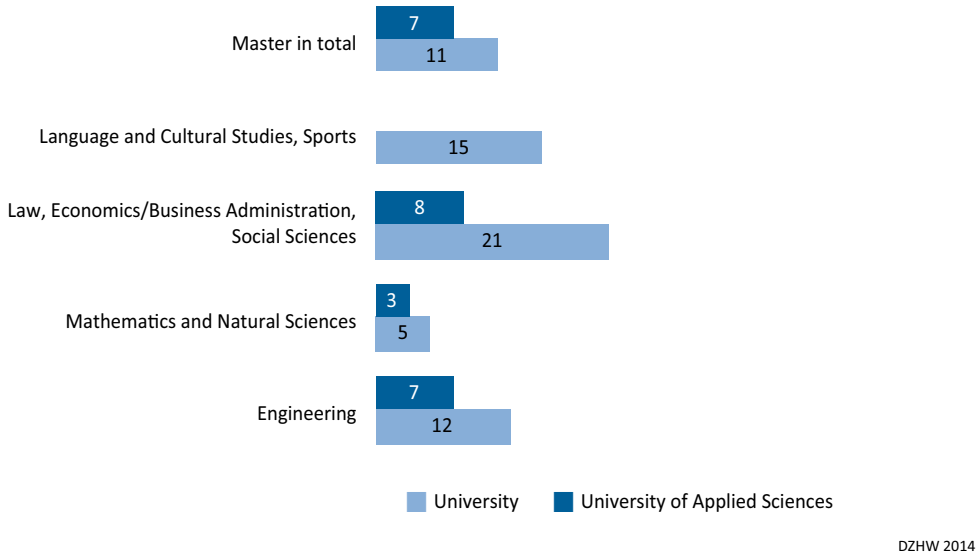

FIGURE 3. Drop-out rates in master degree programmes at universities and universities of applied sciences, by subject groups

roughly at one tenth. The same is true for studies in law where, after a strong dip to 9% for the groups of first-year students from 1992 to 2001, the share of drop outs currently stands at 22%. The below-average numbers of drop-outs in these fields are due to the performance-oriented admission criteria and high motivation and the clear occupational profile of the applicants. Misconceptions about the respective occupational field and the course content are rarely found in these subjects.

Extremely high student drop-out rates of more than 40% are registered for foreign students who obtained their university entrance qualifications abroad — although, this solely applies to students in bachelor programmes. Reliable statements on the reasons for the high numbers of drop-outs in bachelor programmes — as well as the low drop-out figures in master programmes — are not possible because of a lack of empirical research. Hypothetically, it can be assumed that problems between the socialisation of the education experienced in the native countries and the comprehension of teaching and the study requirements at German higher education institutions are responsible for this. In bachelor degree programmes, the higher education institutions seldom organise their first few semesters as a transitional phase for their foreign students.

Development of the Drop-out Theory

The research on higher education in Germany cannot refer to elaborated theories of student drop-out that both meet the multicausal requirements and can claim international validity. However, researchers either developed their own approaches or received appropriate works — especially in the English-speaking world — and tried to make them significant for their own research. Yet the sometimes substantial structural differences between national higher education systems that also affect the analysis of drop-out must be considered. The 'student integration model' developed by Vincent Tinto (Tinto, 1975) in the US in the early 1970s and that

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greatly influenced the discourse on academic success and failure is of great importance in the empirical research on student drop-out. Like the works by Spady (Spady, 1970), Tinto bases himself on the students' social and academic integration. This approach in which drop-out is primarily seen as a process that results from a failed interaction between students as well as between the academic and social system at the higher education institution constitutes the theoretical basis of many studies on student drop-out in Germany (Gesk, 2001). Other approaches drew less attention. For instance, Bean's 'student attrition model' hardly finds any feedback in Germany, even though, unlike Tinto, it put the institutional conditions more sharply into focus. The central aspect of his model is the students' satisfaction with their study situation. According to Bean, satisfaction is linked to several parameters — from students' intellectual developments, grades, and satisfaction with the study content to participation in courses and membership in student organisations (Bean, 1983).

Psychological models have also gained influence in the last few years in the scientific discourse. Individual predispositions, intrinsic study motivation, learning strategies, self-confidence and self-control, social skills and the ability to cope with performance-related pressure are a strong focus of these studies (Schiefele, Streblow & Brinkmann, 2007). In this context, expectancy-value models are also discussed (Hadjar & Becker, 2004). They explain students' performance behaviour based on their expectation of success and their evaluation of the academic degree. While the expectation of success depends on the self-assessment of one's academic skills, the subjective evaluation of academic degrees results from the students' economic and social goals.

Models that explain student drop-out at German higher education institutions with the help of certain individual factors are also considered critical. There is a growing consensus in research on two central aspects. Concerning the problem of multiple causes of student drop-out, all empirical analyses show that there is rarely only one reason for dropping out. Such biography-related decisions are generally due to several reasons that mutually reinforce each other, although different degrees of importance can be assigned to each one (Krieger, 2011; Heublein *et al.*, 2010). There is also a consensus on student drop-out. Drop-out is not understood as a result of spontaneous, short-term decisions, but of a prolonged decision-making and consideration process in which the different influencing factors accumulate in a constellation of problems that makes leaving the higher education institution seem inevitable.

Model of the Drop-out Process

It can no longer suffice to characterise the failed attempt to obtain an academic degree as either individual inability or a problem in efficiency of the higher education system. Hence, the main reasons for dropping out cannot be found in either the students' character or in the higher education institution, e.g. as a consequence of insufficient studying conditions. Rather, it can be assumed that student drop-out is to be understood as being a specific inter-relationship of individual qualifications and institutional conditions. First elements of such an approach can already be found in the research on higher education in the 80s. They were recently strengthened in studies by the DZHW (Heublein *et al.*, 2010). The current project on student drop-out[7] of this research institute aims to describe and analyse the different types of drop-out. It proceeds from the assumption of a

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complex understanding of the drop-out processes. Thus, the decision to drop out, enforced either by the student or the higher education institution, is a consequence of a process that is divided into several phases. In each phase, there are different influencing factors (Figure 4).

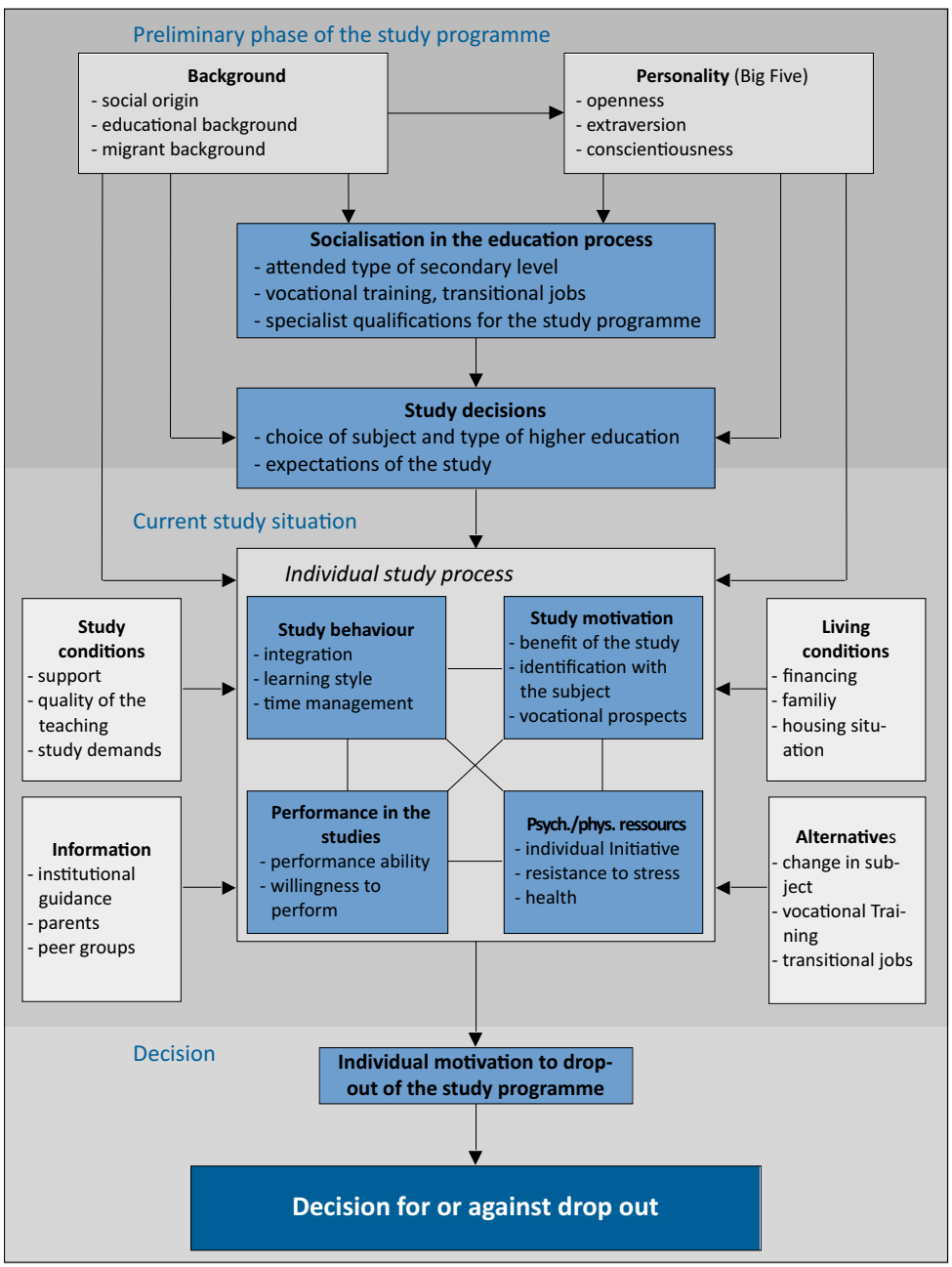

FIGURE 4. Drop out from higher education process

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Some of the determinants that define the drop-out process are already formed in the preliminary phase of the study programme. In this regard, an essential relevance is assigned to social and family origin. Tinto already points to the relevance of origin for social and academic integration that he considers to be central factors of student drop-out (Tinto, 1975). In addition, certain relatively stable personality traits co-determine study behaviour, the dealing with requirements, or the skills in accessing student support. They include individual psychological, cognitive and personal preconditions, as found in the concept of the 'Big Five' (Allport, 1974). Furthermore, great importance must be given to socialisation in the education process before the study programme. The educational choices made prior to enrolment are especially important. They are made on the basis of the expectations regarding the study programme and impact all relevant aspects of the individual study process during studies. The fit between expectations and reality also determines how coherently the study situation is experienced. In these educational choices, there are three fundamental factors: origin, personality and socialisation in the education process.

The second phase of the drop-out process can be found in the study situation itself. It is also a dynamic process that develops from the ongoing interplay between internal and external factors. The internal aspects mark the concrete action of the individual student in the study context, while the external factors refer to those conditions that are set by the institutions. The internal factors that influence academic success include study behaviour (Hadjar & Becker, 2004), study motivation (Schiefele *et al.*, 2007), performance (Hadjar & Becker, 2004; Troche, Mosimann & Rammsayer, 2014) and psychological and physical resources (Brandstätter, Grillich & Farthofer, 2006; Fellenberg & Hannover, 2006). The external factors involve the specific studying conditions, especially study requirements (Blüthmann, Thiel & Wolfgramm, 2011; Rech, 2012), the financing of studies and accommodation (Heublein *et al.*, 2010), the existing alternatives to the current study programme (Hadjar & Becker, 2004) and the supply of information (Heublein *et al.*, 2010). The different factors are composed of several dimensions and characteristics which are mutually dependent. In the course progression, they are subject to modifications that are often due to the relation of the factors amongst themselves. In a successful study programme it is crucial that internal and external factors are coherent despite constant transformations and developments. This means that students must be able to react appropriately to external conditions in their study behaviour and motivations. On the other hand, the study and living conditions and the available information and alternative study programmes must develop in correspondence with the factors of the study process. Academic success is only possible on the basis of mutual correspondence.

The last phase in the study process describes the decision for or against drop-out. If the contradictions between internal and external factors cannot be eliminated, the development of motivations to drop out of the study programme is inevitable.

The model of the drop-out process that was developed by the DZHW is very complex. Since it can also determine varying constellations of problems, it allows for the analysis of the different types of deregistered students, as well as of the various approaches and theoretical perspectives as bases for empirical studies.

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Causes

Despite the lack of an empirically verified and consensual theory of drop-out, there are studies at German higher education institutions on the reasons for premature study termination. However, many refer to intended drop-out (Fellenberg & Hannover, 2006; Multrus, Bargel & Ramm, 2008), but only a few to actual drop-out. While some have a narrow understanding of drop-out, several differentiate between the variations in fluctuation in the higher education system. Most studies on the reasons for dropping out only refer to certain higher education institutions or particular groups of students (Pohlenz, Tinsner & Seyfried, 2007; Blüthmann, Lepa & Thiel, 2008; Thiel *et al.*, 2007; Schiefele *et al.*, 2007). So far, only the DZHW studies have a nationwide representativeness. In 1994, 2000 and 2008 this research institute surveyed a nationwide sample of deregistered students in those years. In comparing these studies, factors vary to a lesser extent than the specific weight with which they influence the cause of drop-out.

Current data on the reasons for student drop-out are still based on the findings of a DZHW survey of deregistered students in the year 2007–2008 in 87 German higher education institutions (Heublein *et al.*, 2010). The drop-outs are a subset of the deregistered students. Among the 4,500 deregistered students who took part in the survey, 2,500 left their programme prematurely. In addition, a representative control group of graduates was recorded. Since the survey took place about six months after the students' deregistration, it cannot be ruled out that some of the test persons who were identified as drop-outs continued their studies at a later stage. Yet, according to their statements, most will not study at a higher education institution again. In the survey, more than 30 motives for dropping out were collected. The respondents were also asked to state the reason that was the most crucial for them. Many individual, institutional and social conditions were collected so that the study broaches the issue of the overall influencing and conditional factors of drop-out in a considerably broader way than many other studies.

According to the respondents, there are three main reasons for dropping out. In each case, about one fifth of the drop-outs refers to individual performance-related problems during the study programme (Figure 5). One student evokes an institu-

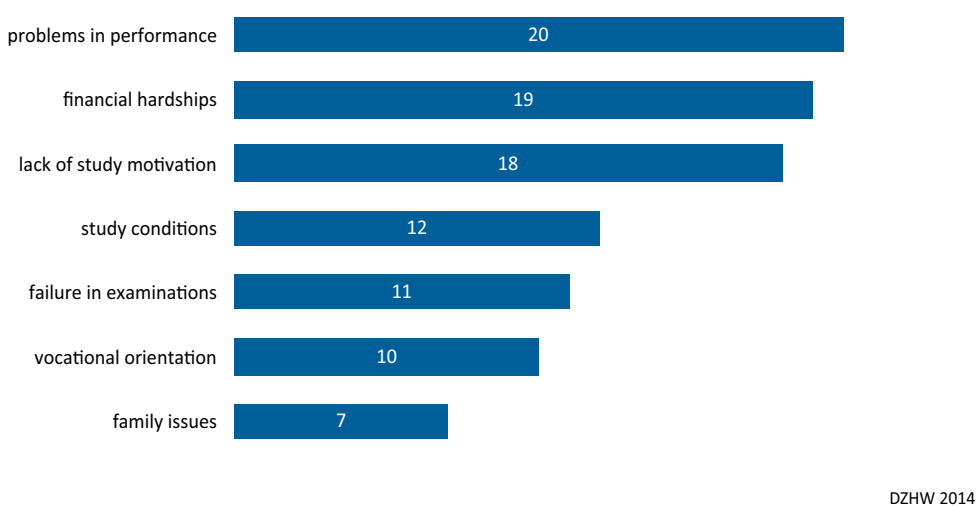

FIGURE 5. Most crucial motives of student drop-out in 2008

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tional dimension: 'the study and assessment load is too extensive' or the 'academic requirements are too high'. Yet, for another, an individual dimension arises if 'doubts on the personal suitability' or the students feel that they 'cannot cope with the performance-related pressure'. Financial problems are also important. The students concerned have problems making their time-consuming job compatible with their academic studies. Moreover, due to a lack of motivation, the programme is frequently left prematurely. The students concerned no longer identify with their subject and its vocational opportunities.

Inadequate studying conditions are also important. In this connection, there is a lack of professional and practical relevance, as well as insufficient organisation of study programmes. Although insufficient studying conditions are often referred to as reasons for dropping out, they are rarely a decisive reason. Another motive is failure in examinations. Students who leave because of failed examinations represent a special group, since, unlike their fellow students who left the higher education institution on their own terms because of their performance, they are convinced of their ability to succeed until the last examination attempt. Some drop-outs abandon their studies because they either were offered a lucrative job or could not imagine practising an academically-oriented job. Women abandon their studies for family reasons. In most cases, they have difficulties in combining childcare or the care of other family members and their studies. 4% of drop-outs decide to leave the higher education institution for health reasons.

The transition to the Bachelor-Master system at German higher education institutions entailed shifts in drop-out motivations. Problems of efficiency and lack of motivation are characteristic of the the bachelor degree programme. Study requirements and a rapid succession of module examinations in the first semesters caused problems for students who started their bachelor programme with insufficient qualifications. Unlike the old study programmes that often did not have 'barriers of selection' before the intermediate exam after four semesters, a comparable effect is already present at the end of the first or second semester in most of the new module exams. Hence, students too seldom succeed in making up for their gaps. Whereas in the old study programmes, drop-outs left the higher education institution after an average of 7 full study semesters, they leave after an average of 2.3 full study semesters in the bachelor programme — at least during the transition period. Hence, the new study structures lead to premature study termination for performance reasons and if students do not identify with the subject and lack motivation.

Dropping out for financial and family reasons has lost its meaning with the introduction of successive study programmes. During the first semesters, students experience certain family problems less often. Financial troubles do not occur at the beginning of the programme, but after several semesters when hardships or conflicts in the reconciliation of their work commitment and academic studies accumulate.

The shift to an earlier study phase is without doubt an outcome of the Bologna Process in German higher education institutions. That the student drop-out rate in bachelor programmes is still above the corresponding rate of the old diploma study programme, however, should not be overlooked. This could suggest that — with the new bachelor requirements — students who initially start their studies with insufficient specialist qualifications fail more frequently than in the past. The former diploma programme provided students with a greater opportunity to

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organise their studies at the beginning of their programme and during the first semesters. This way the higher education institutions were able to lead a larger number of students with insufficient qualifications to academic success on a long-term basis.

With regard to the different institutional and individual conditions that influence drop-out and its probability, the German research on higher education shows many valid results. They are partially based on a comparison of the survey results between drop-outs and graduates (Heublein *et al.*, 2010).

Doubts, especially concerning the choice of the field of studies and enrolment in a programme that is not the initial choice, considerably increase the risk of drop-out. Motivation is also important. A solely or primarily extrinsic motivation (e.g. a high income or good career opportunities) proves to be unsustainable for a successful study programme if it is not supported by a close identification with the subject. Since the bachelor programmes are frequently subject to local admission restrictions, there are indications of higher shares of first-year students who do not study in their preferred subject.

Insufficient educational qualifications for a programme pose a risk for the acquisition of a degree. Gaps in mathematics, natural science and business administration increase the risk of failure. These often seem to be the consequence of the choice of specialised course or core study area in upper secondary education that does not correspond to the chosen subject in higher education. The type of school attended also plays a prominent role. Students who obtained their admissions qualification at a Fachgymnasium, a Kolleg or a Fachoberschule currently fail more often than their fellow students who obtained their general higher education entrance qualification.

Compared to successful students, students who dropped out of the higher education institution, most notably in the bachelor degree programme, feel overwhelmed by the amount of material, the level of specialist requirements and the autonomy that is necessary for the organisation of their studies. Hence, they evaluate their achievements more critically than graduates. Study conditions — e.g. curriculum, clarity about the performance requirements, organisation of the studies, teaching quality — are consistently evaluated more critically by drop-outs than by successful students. On another note, drop-outs are less satisfied with student support, especially in natural and engineering sciences. The equipments of the higher education institution do not seem to influence the decision to drop out. The dialectical process in which the studying conditions unfold should, however, not be misjudged in the evaluation of this finding. Student support and some of the studying conditions are not provided from the start, but have to be accessed by the students who must be able to draw a conclusion regarding their level of performance and, more importantly, their gaps. In a next step, they must check the offers at the higher education institution that are suitable for their situation. Ultimately, communication abilities are also necessary. This means that the drop-outs' critical opinion about studying conditions and student support does not only reveal a lack of offerings of specific support possibilities, but an incapacity to access relevant offers. These outcomes correspond to the findings on the social integration of drop-outs. Strong social interconnections decrease the risk of dropping out. Students who seek social contacts outside the higher education institution rather than engage in study groups prove to be more at risk of dropping out of the higher education system.

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Type of drop-outs I	Type of drop-outs II	Type of drop-outs III
Preliminary phase of the programme: - educational deficits/ poor grade point average - poor knowledge about the study demands - extrinsic choice of subject	Preliminary phase of the programme: - wrong expectations/ study content/occupational profile - not the favoured subject - extrinsic choice of subject	Preliminary phase of the programme: - financing of the studies is unclear - long transition to the degree programme
Study situation: - insufficient student support - didactical deficiencies	Study situation: - no motivational support - missing practical relevance	Study situation: - financing through the German federal Law on Support in Education/ occupation - intensive occupation
Study situation: - exessive specialist demand - failure in examinations	Study situation: - decreasing interest in the subject - decreasing willingness to perform - doubts on the study suitability	Study situation: - lack of integration
Drop-out due to performance-related reasons	Drop-out due to motivational reasons/ change in vocational direction	Drop-out due to financial reasons

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FIGURE 6. Process of drop-out from higher education

The main causes for students to dropout also include their financial situation and occupation during studies. Problems related to the financing of the studies are the second most important reason for students to drop out. Drop-outs depict their financial situation during their studies far more critically than graduates. They receive less financial support from their parents and are more frequently dependent on public funding. An occupation during studies affects academic success since it takes up a great deal of their time. The great significance that is attached to financial reasons for dropping out indicates an influence of the socio-economic background: students from lower-income families are at a higher risk of dropping out.

In the empirical analysis, three main at-risk groups can by identified on the basis of these findings (Figure 6). The first includes students who start their studies with educational gaps and a poor grade point average in upper secondary education. Their decision is to a great extent determined extrinsically. If, in addition, they do not receive any motivational support and are not able to seek support from teaching staff as well as from other students, they quickly face problems to fulfil their study tasks. Another at-risk group starts their studies at a higher education institution with false expectations with regard to the specialist contents and career prospects. Their decision is marked by uncertainty or possibly even by certainty that they are not studying in their preferred subject. Interest in the subject and the willingness to perform decrease in the first semesters. The students of the third at-risk group invest more time in a job during their studies than their fellow students. From their point of view, this is the only way of financing their studies. This group often includes students who completed a vocational training prior to their studies or whose transition to the degree programme took a long time. If the job exceeds a certain time-scale, a collision with the study demands can be foreseen. These students have an above average risk of dropping out for financial reasons.

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Preventive Measures to Increase Academic Success

The high drop-out rates in the bachelor degree programme obliged the German higher education institutions to legitimise in connection with the growing demand of the economy for academically-trained specialists. They increasingly face requests to guide more first-year students towards a completed academic degree. Academic success and drop-out are understood to be an interrelationship between individual qualifications and the manner of studying on the one hand and conditions at the higher education institution and in the social environment on the other.

In this context, it is not surprising that academic success initiatives have been developed by many higher education institutions in the last few years. Many are realised within the framework of a programme of the Federal Government and the state, the 'Quality Pact for Teaching'. 186 German higher education institutions have engaged in this funding programme that is set to run until 2020. The improvement of the orientation phase plays a special role. With the transition to the bachelor degree programme, many problems arose with the reshaping of the curricula. The increasing heterogeneity of the first-year students has also contributed to these problems. In addition to the issues about the fit of after-school preparation and study demands in the first few semesters, problems concerning the study programmes — too much material, too many examinations, too high demands at the start of the study programme already — began to appear. Several higher education institutions try to avoid such problems by offering better information to student applicants, reforming the curricula, offering bridging courses to compensate specialist gaps, and offering mentors and tutors for the new first-year students in the first semesters. The 'Verbesserte Lehrqualität' project of the University of Würzburg can be mentioned as an example of the activities that are supported within the framework of the 'Quality Pact for Teaching'. It offers applicants an online test about their interests that was developed by the psychological consultancy of the university. For the transition to the higher education institution, propaedeutic courses are offered not only in natural sciences, but also in humanistic subjects. Individual student counselling, as well as an introduction to the offerings of the university are integrated in these courses. The tutoring and mentoring programme for first-year students was expanded. The project also includes a 'Studierwerkstatt' which comprises interdisciplinary tools for a successful course progression.

For the applicants, the study decisions are increasingly complex because of the multiplicity and inconsistency of the criteria and aspects to be considered. The massive change that is currently taking place in the German higher education system because of a stronger vertical distinction between the higher education institutions also contributes to this. Often, the decisions are irrational and lead to wrong decisions. Hence, many quality assurance measures focus on the admission to higher education. The higher education institutions try to increase the certainty of the choice of study through self-assessments and specific selection procedures. This seems to succeed, especially if these procedures include consultative and diagnostic functions. Hence, self-assessment systems are available for every subject in North Rhine-Westphalia.

An important step towards further developments of the bachelor degree programmes to increase academic success has been taken by some higher education

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institutions in Baden-Wuerttemberg. With the help of funding from the federal state, all too rigid study guidelines and curricula are loosened by offering flexible course progressions at an individual pace. The higher education institutions thus react to the increasing heterogeneity of their first-year students and their different background knowledge, interests and talents. In addition, the projects envisage the introduction of part-time studies, successive study models, or even a polyvalent first semester that enhances the study decision.

These measures that have developed in the last two to three years will succeed in reducing drop-out rates. The question, however, remains whether an improvement in quality is possible. The strategies for supporting academic success concentrate on certain types of drop-outs only. They do not take into account the empirical finding that there is not one dominant type of drop-out, but several that are of similar importance. Financial problems that influence drop-out were rarely emphasised in assistance measures, for instance. The overarching strategies and the funding programme on academic success do not sufficiently reflect drop-out as a process that includes several stages. There is a need for funding and assistance to school education, the transition to the higher education institution and studies. Problems that lead to drop-out develop in secondary school, build up during the transition period and become even more acute in the higher education institution, so that the decision to abandon one's studies is inevitable. An overarching and long-term agreement on the different educational stages that aims at minimising risks will not only reduce the discrepancies between secondary school, vocational training and higher education, but will also open the way to academic success for all applicants who have the necessary cognitive abilities.

Ulrich Heublein, Deutsches Zentrum für Hochschul- und Wissenschaftsforschung, Goseriede 9, 30159 Hannover, Germany, heublein@dzhw.eu

NOTES

1. This position is held by the VDMA (Verband Deutscher Maschinen- und Anlagenbau e.V.), the largest German engineering industry network. See also VDMA (Hrsg.) (2013). *Maschinenhaus — Campus für Ingenieure. Die VDMA-Initiative für Studienerfolg* (Frankfurt a. M.).
2. An exception is drop-outs from master degree programmes because they already hold a first academic degree in the form of their bachelor degree.
3. Data protection concerns play an especially important part. In some countries such as Switzerland or Austria, such statistics of course progressions can be created.
4. The efforts in the framework of the National Educational Panel Study, however, point to an improvement in the research situation.
5. Programmes that prepare for a teaching profession and lead to a Master of Education are not included in the calculation of drop-out in master degree programmes.
6. This is documented by current graduate surveys, e.g. Rehn, T.; Brandt, G.; Fabian, G.; Briedis, K. (2011) *Hochschulabschlüsse im Umbruch. Studium und Übergang von Absolventinnen und Absolventen reformierter und traditioneller Studiengänge des Jahrgangs 2009* (Hannover: HIS, Forum Hochschule 17|2011).

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7. The current DZHW project on the causes of drop-out started at the beginning of 2014 and will last until mid-2016. It includes multiple modules, from the determination of drop-out rates to the survey of drop-outs and the analysis of preventative and intervening measures in different faculties. First findings are expected in 2015.

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